Data Preparation and Optimization of a k-Nearest Neighbors Classifier

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I. Introduction

A. Objective

The objective of this study is to clean and transform a noisy customer dataset in order to maximize the predictive performance of a k-Nearest Neighbors (kNN) classifier. The model aims to predict **Purchase_Likelihood** based on customer demographic, behavioral, and transactional attributes. Because kNN is a distance-based algorithm, its performance is highly sensitive to data quality, feature scaling, and noise, making data preparation a critical step in the modeling process.

B. Dataset Overview and Identified Noise

An initial manual inspection of the *noisy_knn_300.csv* dataset revealed several data quality issues that justified the need for a structured preparation pipeline:

- **Inconsistency:** The *Gender* attribute appeared in multiple formats (e.g., M, male, Female, fem), representing the same information using different textual conventions.
- **Outliers:** The *Age* column contained physically impossible values (e.g., negative ages and values greater than 100), which could distort distance calculations.
- **Redundancy:** The columns *Items_Purchased* and *Purchase_Volume* conveyed nearly identical information, with the latter being a categorical interpretation of the former.
- **Missingness:** Several columns, including *Age*, *Annual_Income*, and *Total_Spend*, contained missing (NaN) values.
- **Scale Imbalance:** *Annual_Income* had values in the tens or hundreds of thousands, while *Support_Calls* ranged only from small integers. Without scaling, features with larger ranges would dominate distance computations in kNN.

These issues showed that the dataset needed to be cleaned and transformed before applying any machine learning model.

II. Methodology (Phases 2 and 3

A. Data Preparation Pipeline

The data preparation process was implemented in Google Colab using a step-by-step and modular approach. Each task was placed in a separate code cell and explained using Markdown. The main steps in the pipeline were:

1. **Removing Duplicates:** Duplicate records were removed using the *Customer_ID* column to ensure that each row represented a unique customer.
2. **Handling Outliers:** Invalid values in *Age* and *Annual_Income* were replaced with missing values and then filled using the median.
3. **Handling Missing Values:** Remaining missing values in numeric columns were filled using median imputation to avoid losing too much data.
4. **Standardizing Categorical Data:** Inconsistent text values in columns such as *Gender* and *Device_Type* were cleaned and standardized.
5. **Feature Engineering:** A new categorical feature called *Purchase_Volume* was created from *Items_Purchased*, and tax rate information was added based on *Store_Region*.
6. **Feature Scaling:** Min–Max scaling was applied to all numeric attributes so that they would be on the same scale.
7. **Encoding and Column Removal:** Categorical variables were converted using one-hot encoding, and unnecessary columns like *Customer_ID* and *Purchase_Date* were removed.

These steps resulted in a clean and structured dataset that was suitable for training a kNN model.

B. Experimental Setup

Two experiments were conducted to evaluate the impact of data preparation:

- **Baseline Run:** The raw dataset was reduced to numeric columns only, rows with missing values were removed, and a kNN model with $k = 5$ was trained.
- **Optimized Run:** The cleaned dataset produced by the full data preparation pipeline was used to train another kNN model with $k = 5$.

Both experiments used an 80/20 train-test split, and model performance was evaluated using accuracy and confusion matrices. The value of k was also tested from 1 to 20 to determine the optimal number of neighbors.

III. Results and Analysis

A. Data Quality Comparison

Before data preparation, the dataset contained missing values, inconsistent formats, outliers, and unscaled features. After applying the preparation pipeline, the dataset became more consistent, complete, and properly scaled. These improvements made the data more suitable for distance-based learning algorithms such as kNN.

B. Performance Comparison

Table I shows the comparison between the baseline and optimized kNN models.

Table I. Baseline vs. Optimized kNN Performance

Model Run	*Accuracy*	*Confusion Matrix*
Baseline	42.86%	[[3 1] [7 3]]
Optimized	68.97%	[[22 9] [9 18]]

The optimized model achieved better accuracy compared to the baseline model, indicating that proper data preparation improved the model's performance.

C. k-Parameter Optimization

The optimized model was tested with values of *k* ranging from 1 to 20. The results showed that very small values of *k* led to unstable predictions, while very large values reduced accuracy. The highest accuracy was observed at **k = (3)**, which was considered the optimal value.

IV. Discussion and Conclusions

One of the main challenges in this activity was dealing with noisy and inconsistent data without removing too many records. This was addressed by using median imputation and standardization techniques. Another challenge was preparing categorical data in a way that could be used by the kNN algorithm, which was solved through proper encoding and scaling.

This activity highlights the importance of data preparation in machine learning. The results show that cleaning, scaling, and structuring the dataset significantly improved the performance of the kNN classifier. Overall, better data quality leads to better and more reliable model predictions.